

Figure 9. Example analysis: Bivariate distribution - Providence second grade reading

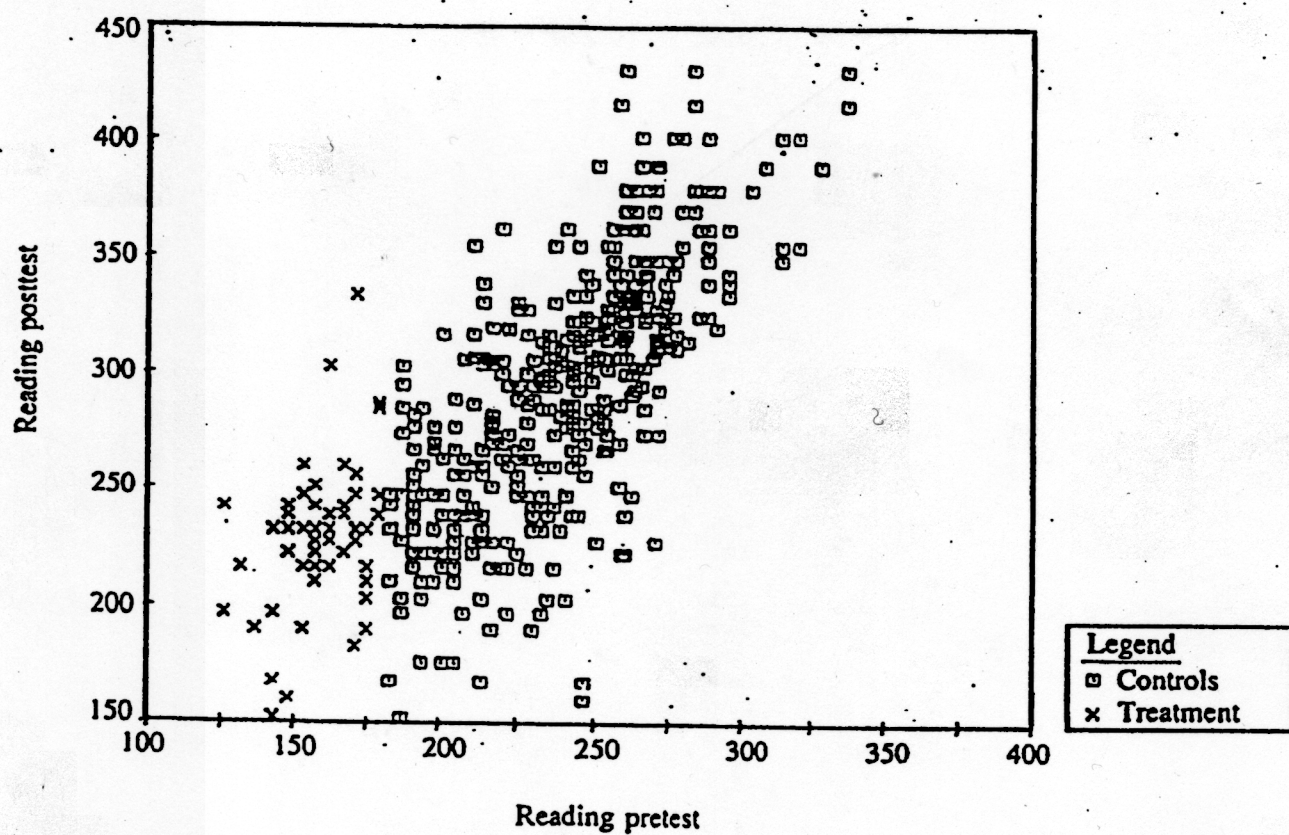


Table 1. Estimates for initial model, second-grade reading program, Providence, RI school district, 1978-79

Variable	b	SE(b)	p
Constant	216.27346	9.02495	<.001
Linear (x_i)	.88212	.49993	.078
Program effect (Z_i)	21.69478	17.24679	.209
Linear interaction ($x_i z_i$)	-2.52099	2.85144	.377
Quadratic (x_i^2)	.00816	.00774	.292
Quadratic interaction ($x_i^2 z_i$)	-.13967	.13471	.300
Cubic (x_i^3)	-.00003	.00003	.266
Cubic interaction ($x_i^3 z_i$)	-.00171	.00173	.324
$R^2 = .56919$			

Table 2. Estimates for revised models, second-grade reading program, Providence, RI school district, 1978-79

Variable	b	SE(b)	p
Revision 1: cubic terms eliminated			
Constant	209.1262	6.3448	<.001
Linear (x_i)	1.3919	.2006	<.001
Program effect (Z_i)	37.4535	13.5511	.006
Linear interaction ($x_i z_i$)	-.4602	1.2013	.702
Quadratic (x_i^2)	-.0003	.0014	.836
Quadratic interaction ($x_i^2 z_i$)	-.0023	.0246	.925
$R^2 = .56706$			
Revision 2: cubic and quadratic terms eliminated			
Constant	210.11	4.22	<.001
Linear (x_i)	1.35	.06	<.001
Program effect (Z_i)	35.84	10.06	<.001
Linear interaction ($x_i z_i$)	-.51	.39	.193
$R^2 = .56702$			
Revision 3: linear term only			
Constant	210.87	4.18	<.001
Linear (x_i)	1.34	.06	<.001
Program effect (Z_i)	44.61	7.50	<.001
$R^2 = .56542$			